

Emanuele Mule

Paris, France - Research Engineer at Télécom Paris, Intitute Polytechnique de Paris
emi.mule2001@gmail.com - [LinkedIn Profile](#) - [GitHub Profile](#) - [Website](#) - [Google Scholar](#)

WORK EXPERIENCE

Research Engineer

Télécom Paris, Institut Polytechnique de Paris, Paris, France

September 2026 – Present

- Researching graph neural networks and higher-order structures, focusing on **inferring latent hypergraphs** of group interactions from data without relation labels.
- In parallel, working on **faster Graph Fourier Transforms** using low-rank representations and one- and two-step rank updates.

AI Engineer

Hermes Trade, Florence, Italy

January 2026 - July 2026

- Research and development of methods to enhance the performance of **embedding models for retrieval tasks**.
- Design and implementation of an automated pipeline for **unsupervised text classification**.

Research Assistant

Sapienza University of Rome, Rome, Italy

April 2025 - January 2026

- Conducted research for a **publication on Sheaf Neural Networks and Directed Hypergraphs** with RSTLess @ Sapienza under supervision of Prof. Silvestri.
- Proposed a **complex-valued Laplacian operator** for directed hypergraphs, whose integration into convolutional architectures yielded substantial performance gains over existing methods.
- Accepted at ICLR 2026 as a main conference paper.
- Paper: [Directional Sheaf Hypergraph Networks: Unifying Learning on Directed and Undirected Hypergraphs](#)

Research Assistant

Sapienza University of Rome, Rome, Italy

June 2024 - February 2025

- Conducted research for a **publication on Ground-to-Aerial Image Matching** with ALCOR Lab @ Sapienza under supervision of Prof. Amerini.
- Trained a **deep learning model employing Semantic Segmentation features** along RGB images to match accurately a query ground-view image to a satellite one.
- Accepted at WACV 2025 Workshop on AI for Multimedia Forensics & Disinformation Detection Workshop.
- Paper: [Enhancing Ground-to-Aerial Image Matching for Visual Misinformation Detection Using Semantic Segmentation](#)

Teaching Assistant

Sapienza University of Rome, Rome, Italy

September 2022 - June 2023

- Development of **course materials** for lectures and lab assignments.
- Conduction of **lab sessions** and **office hours** for the courses "Fundamentals of Programming" and "Programming Methodologies".

EDUCATION

Sapienza University of Rome

Rome, Italy

Master in Artificial Intelligence and Robotics

September 2023 - October 2025

- Final Grade: **110 cum laude/110**
- Final GPA: **29.5/30**
- In the Top **1%** students of the course, based on GPA, according to "Bonus Esami" certification for the first and second year

Sapienza University of Rome

Rome, Italy

Bachelor in Computer and Control Engineering

September 2020 - July 2023

- Final Grade: **110 cum laude/110**
- Final GPA: **28.98/30**
- In the Top **1%** students of the course, based on GPA, according to "Bonus Esami" for second and third year

MAIN TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, MATLAB, JavaScript

Libraries and Tools: PyTorch / Lightning, TensorFlow, Sklearn, Pandas, Numpy, OpenCV, Git, Node.js, ROS

Databases Querying and Management: SQL, MongoDB

Topics: Artificial Intelligence, Reinforcement Learning, Deep Learning, Robotics, NLP, Computer Vision

LANGUAGE SKILLS

English: Fluent speaker | **Italian:** Native Speaker | **Albanian:** Native Speaker

EXTRACURRICULAR ACTIVITIES

- **ESSAI2025 - European Summer School on Artificial Intelligence** in Bratislava - June 2025
- **ITASEC22 - Italian Conference on CyberSecurity** in Rome - June 2022